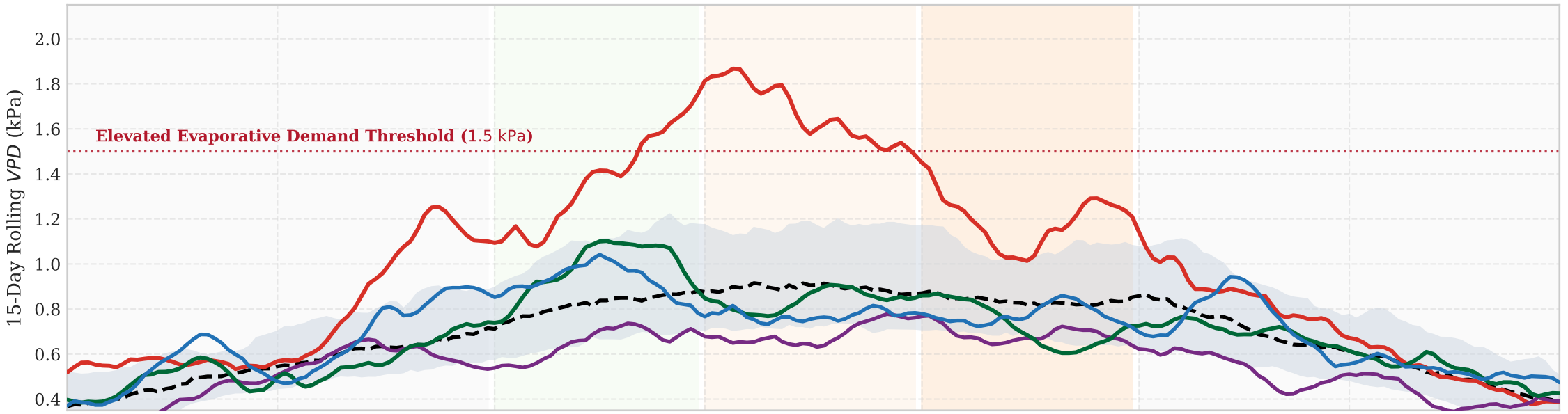
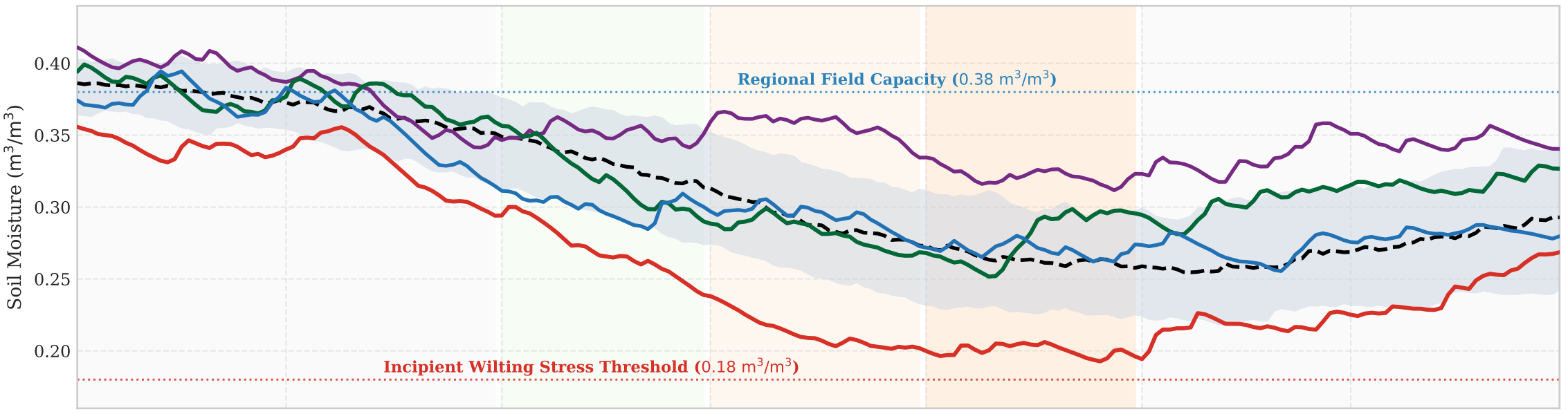
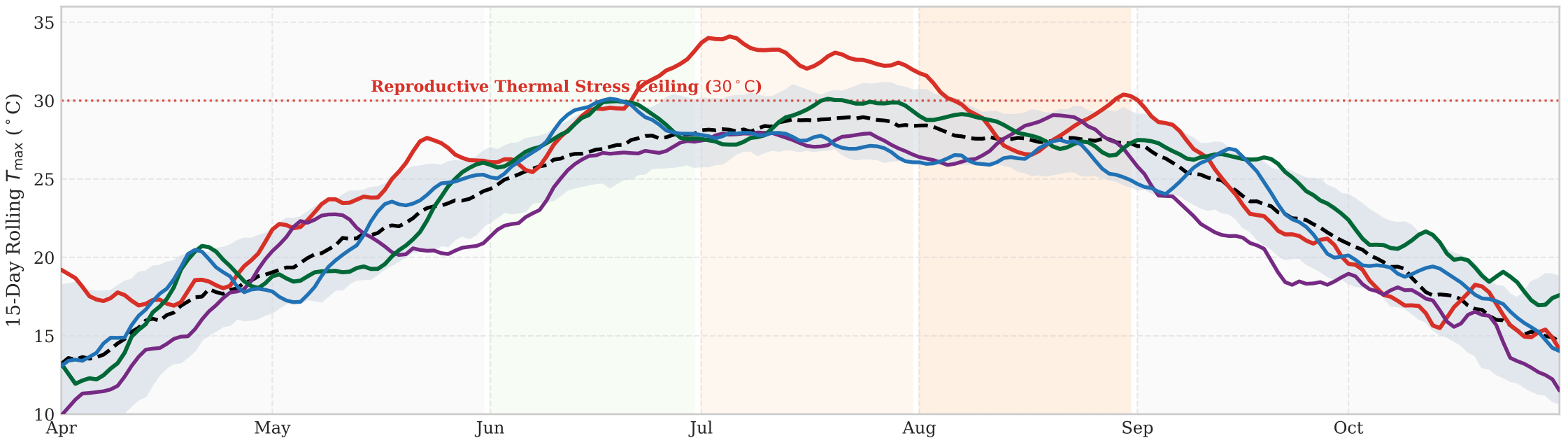
(a) Atmospheric Evaporative Demand (15-Day Rolling *VPD*)



(b) Integrated Root-Zone Soil Moisture Dynamics (SM_{root}, 0–100 cm)



(c) Thermal Regimes (15-Day Rolling Maximum Temperature)



Climatological Range (10th–90th %ile)
Climatological Median
2012 Shock: Flash Drought (-11.7%)
1993 Shock: Midwest Flood (-10.5%)
2016 Bumper: Optimum Buffer (+10.8%)
1994 Bumper: Ideal Balance (+10.7%)